

BLOOD DONATION MANAGEMENT SYSTEM

A MINI-PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

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This mini project report is submitted for the viva voce examination to be held on

INTERNAL EXAMINER _____

EXTERNAL EXAMINER _____

ABSTRACT

In the existing system, blood bank operations often suffer from manual record-keeping, leading to issues like slow retrieval of donor and patient data, difficulty in real-time inventory tracking, and potential human errors in critical processes, which poses a significant risk to the timely and safe supply of blood components. To mitigate these risks and enhance operational efficiency, we have developed a Web-based Blood Bank Management System (BBMS) utilizing a robust database framework; the primary goal of this system is to provide a centralized, automated, and error-free platform to manage the complete donation lifecycle, including systematic donor screening, real-time blood stock inventory management, and accurate issuance/cross-matching records. The implementation of this system significantly reduces paperwork and minimizes data entry errors, enabling instantaneous access to critical data and ultimately ensuring the safer, faster, and more effective distribution of blood components, thereby improving patient safety outcomes. The system is developed using Java Swing for the front-end interface and MySQL as the back-end database, connected through the JDBC (Java Database Connectivity) driver.

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CHAPTER 1

INTRODUCTION

The Blood Donation Management System is a computerized database application developed to manage the various aspects of blood donation and distribution efficiently. Blood is one of the most vital components for saving lives, and maintaining an organized system to handle donor information, blood inventory, and recipient details is essential for hospitals and blood banks. Traditional manual record-keeping methods are time-consuming, prone to errors, and make it difficult to retrieve information quickly in emergency situations.

The Blood Donation Management System is designed to simplify and automate the management of blood donation activities within hospitals, clinics, and blood banks. The scope of this project includes the development of a relational database that efficiently stores, retrieves, and manages donor, recipient, and blood inventory information.

In many areas, especially in developing regions, blood donation and availability services are still managed through outdated manual processes or fragmented local systems. This leads to delays in finding suitable donors, mismatches in blood group availability, and difficulties in maintaining proper records.

The main aim of the Blood Donation Management System is to design and develop a database-driven application that efficiently manages blood donor information, blood stock details, and recipient records. The system seeks to ensure quick access to accurate data, improve coordination between donors and blood banks, and enhance the overall efficiency of blood donation and distribution processes.

CHAPTER 2

SYSTEMSPECIFICATIONS

Hardware Specifications:

Processor: Intel i5

RAM: 16 GB

Hard Disk: 500 GB

Software Specifications:

Operating System: Windows 11

Front-End: Java Script

Back-End: MySQL and JDBC(Java Database Connectivity)

Language: Java

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MODULE DESCRIPTION

3.1 Module Description

The Blood Donation Management System is divided into several modules that handle different operations such as managing donors, recipients, blood stock, and donation camps. Each module performs a specific function to ensure smooth operation and efficient data management in the system.

3.1.1 Admin Login Module

This module allows the administrator to log into the system using valid credentials. The admin can manage donor and recipient records, update blood stock, approve blood requests, and generate reports.

3.1.2 Donor Management Module

This module manages all donor-related information. It allows adding, updating, and deleting donor details and keeps track of donors' blood groups, contact details, and last donation dates.

3.1.3 Recipient Management Module

This module maintains records of recipients who request blood. It stores details such as patient name, required blood group, and hospital information, helping match requests with available donors or blood stock.

3.1.4 Blood Inventory Module

This module handles the management of available blood units in the blood bank. It updates stock after donations or issues, checks for low stock levels, and ensures that each blood unit's expiry date is monitored.

3.1.5 Blood Donation Camp Module

This module is used to organize and manage blood donation camps. It stores details about camp dates, venues, and organizers, and keeps records of donors who participated in each camp.

3.1.6 Request and Issue Module

This module processes blood requests from hospitals or patients and records the issuing of blood units. It helps track pending and completed requests and ensures accurate delivery.

3.1.7 Database Module

The database module stores and manages all system data securely. It connects the different modules, ensuring that donor, recipient, and stock information can be accessed and updated efficiently.

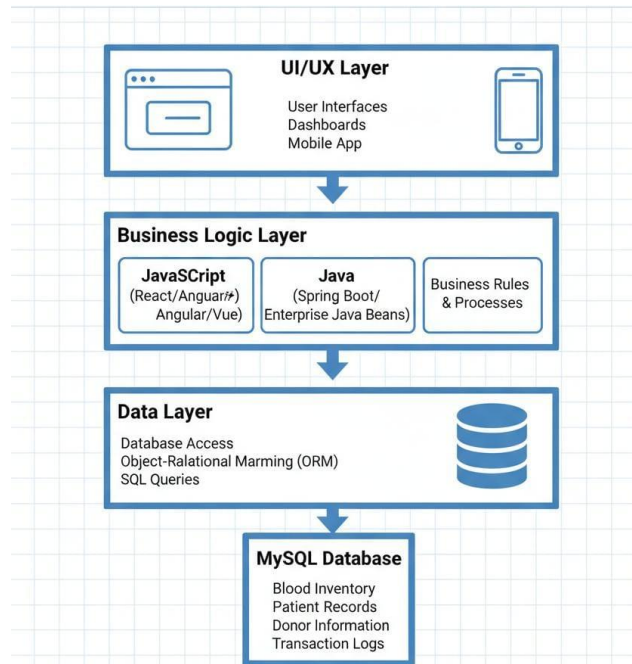
3.1.8 Report Generation Module

This module generates detailed reports for administrators. Reports include donor lists, stock availability, donation history, and summary statistics, assisting in effective decision-making and analysis.

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ARCHITECTURE DIAGRAM & ER DIAGRAM

4.1 ARCHITECTURE DIAGRAM



4.1 Architecture Overview

The Blood Bank Management System also follows a three-tier architecture, consisting of the Presentation Layer, Business Logic Layer, and Data Layer. This structured approach significantly enhances maintainability, scalability, and efficiency by separating concerns related to user interaction, core logic processing, and data management into distinct, interconnected layers.

1. Presentation Layer

The Presentation Layer is the user-facing component of the system. It provides intuitive interfaces for various users, including blood bank staff, administrators, and potentially even donors or recipients. This layer enables users to interact with the application through screens designed for donor registration, blood inventory management, request processing, and report generation. Implemented using modern web technologies such as JavaScript.

2. Business Logic Layer

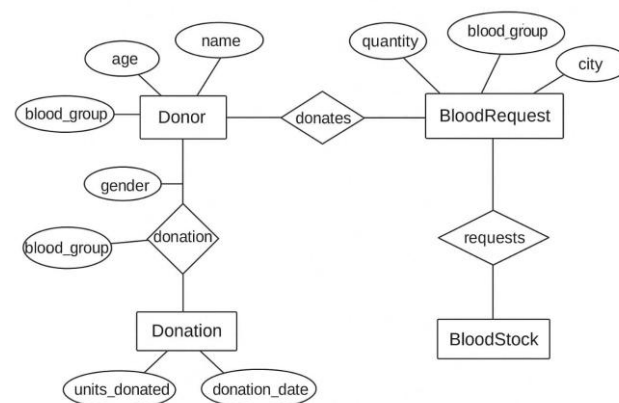
The Business Logic Layer (BLL) forms the core intelligence of the Blood Bank Management System. It is responsible for handling all critical processing, decision-making, and application of business rules. This layer validates inputs received from the Presentation Layer, enforces system policies, and coordinates complex operations between the user interface and the database. Key functionalities include verifying donor eligibility, managing blood product inventory levels, matching blood requests with available units, processing blood donations and transfusions, and generating alerts for low stock. This layer is often implemented using robust server-side

languages like Java (with frameworks such as Spring Boot or Enterprise Java Beans), ensuring data accuracy, adherence to medical protocols, and the enforcement of all business rules before any data interaction.

3. Data Layer

The Data Layer is dedicated to the persistent storage, efficient retrieval, and comprehensive management of all data within the system. It utilizes a robust relational database, typically MySQL, to maintain a structured collection of tables for blood inventory, donor records, patient information, transfusion history, staff details, and transaction logs. All data manipulation operations, such as adding new donor information, updating blood unit statuses, or deleting outdated records, are executed through optimized SQL queries using appropriate database connectivity mechanisms. This layer is crucial for ensuring data integrity, security, and consistency across the entire Blood Bank Management System, supporting reliable operations and accurate reporting.

4.2 ENTITY-RELATIONSHIP



Show how each entity and relationship is converted into tables.

Example:

STUDENT(Student_ID, Name, Dept, Email)

COURSE(Course_ID, Course_Name, Credits)

ENROLLMENT(Student_ID, Course_ID, Grade)

4.3 RELATIONAL ALGEBRA EXPRESSIONS

DEFINITION

Relational Algebra is a formal language used to retrieve and manipulate data stored in relational databases. It uses mathematical operations like select, project, join, union, and rename to perform queries on tables.

Examples (Blood Donation Management System)

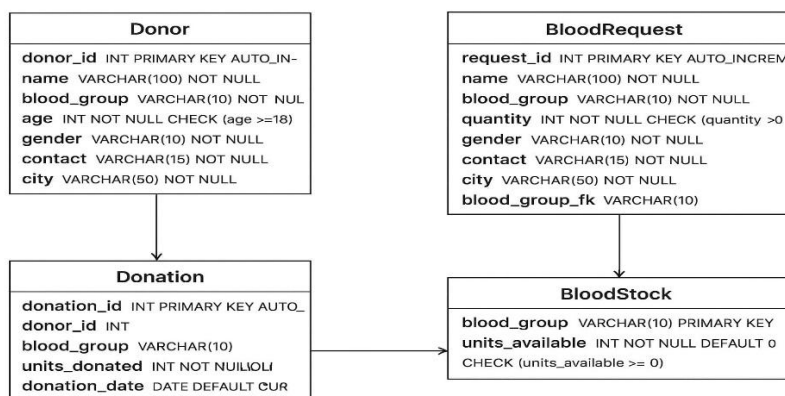
1. Find all donors with blood group 'A+'

σ Group_Name = 'A+' (BLOOD_GROUP $\bowtie$ DONOR)
 (Select donors whose blood group is A+)

2. List the names of donors from Chennai
 π Name (σ City = 'Chennai' (DONOR))
 (Project only the Name column after selecting Chennai donors)

3. Show all pending blood requests
 σ Status = 'Pending' (REQUEST)

4.4 RELATIONAL SCHEMA DIAGRAM



4.5 CONSTRAINTS USED IN THE PROJECT ,
 In the Blood Donation Database Management System, various database constraints are applied to maintain data integrity, accuracy, and consistency across all tables. The commonly used constraints are PRIMARY KEY Constraint ,FOREIGN KEY Constraint, NOT NULL Constraint, CHECK Constraint ,UNIQUE Constraint.

Table Name	Column Name	Data Type	Constraint
AVAILABLE DONORS	NAME	VARCHAR2(20)	NOT NULL
1	AGE	INT	CHECK (AGE >= 18)
2	GENDER	VARCHAR2(5)	CHECK (GENDER IN ('Male', 'Female', 'Other'))
3	CITY	VARCHAR2(15)	NOT NULL
4	LAST DONATION	DATE	CHECK (LAST_DONATION <= SYSDATE)

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DATABASE IMPLEMENTATION

5.1 DATABASE CONNECTIVITY CODE WITH JAVA:

```
package com.bloodbank.dao;

import java.sql.*;

public class DBConnection {

    private static Connection conn = null;

    public static Connection getConnection() {

        try {

            if (conn == null) {

                Class.forName("com.mysql.cj.jdbc.Driver");

                conn = DriverManager.getConnection(

                    "jdbc:mysql://localhost:3306/bloodbank", "root", "password");

            }

        } catch (Exception e) {

            e.printStackTrace();

        }

        return conn;

    }

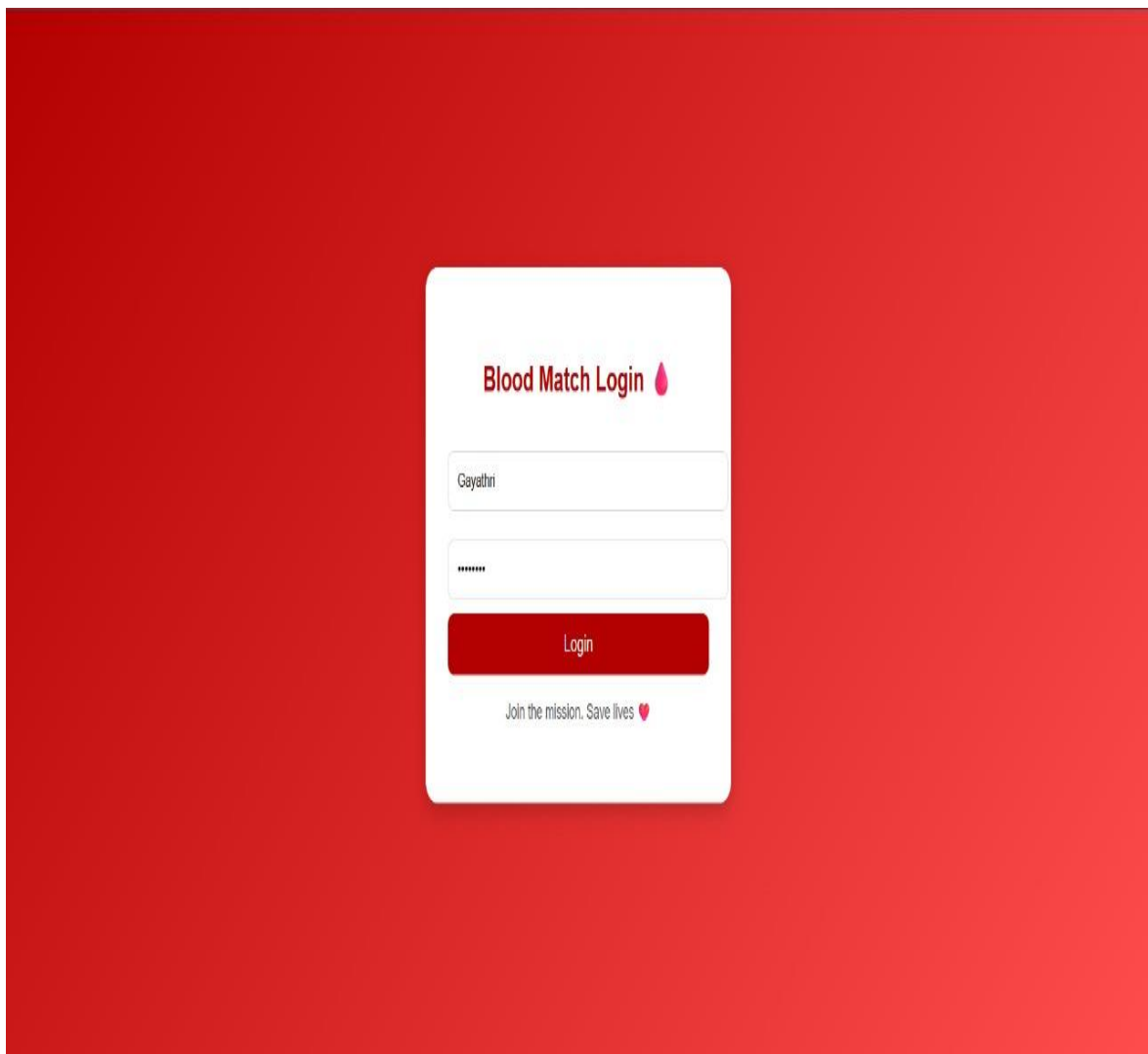
}
```

This Java class **DBConnection** is used to create and manage a connection to a **MySQL database**. It loads the MySQL JDBC driver (`com.mysql.cj.jdbc.Driver`). Connects to the database **"bloodbank"** on **localhost** using the username **root** and password **password**. It uses a **static connection** so the same connection can be reused throughout the application. If any error occurs, it prints the error details.

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SCREENSHOTS

LOGIN PAGE : This is the secure **entry point** for users (likely administrators) to access the system. It ensures that only registered individuals can participate in the Blood Match Network.(User Based Access)

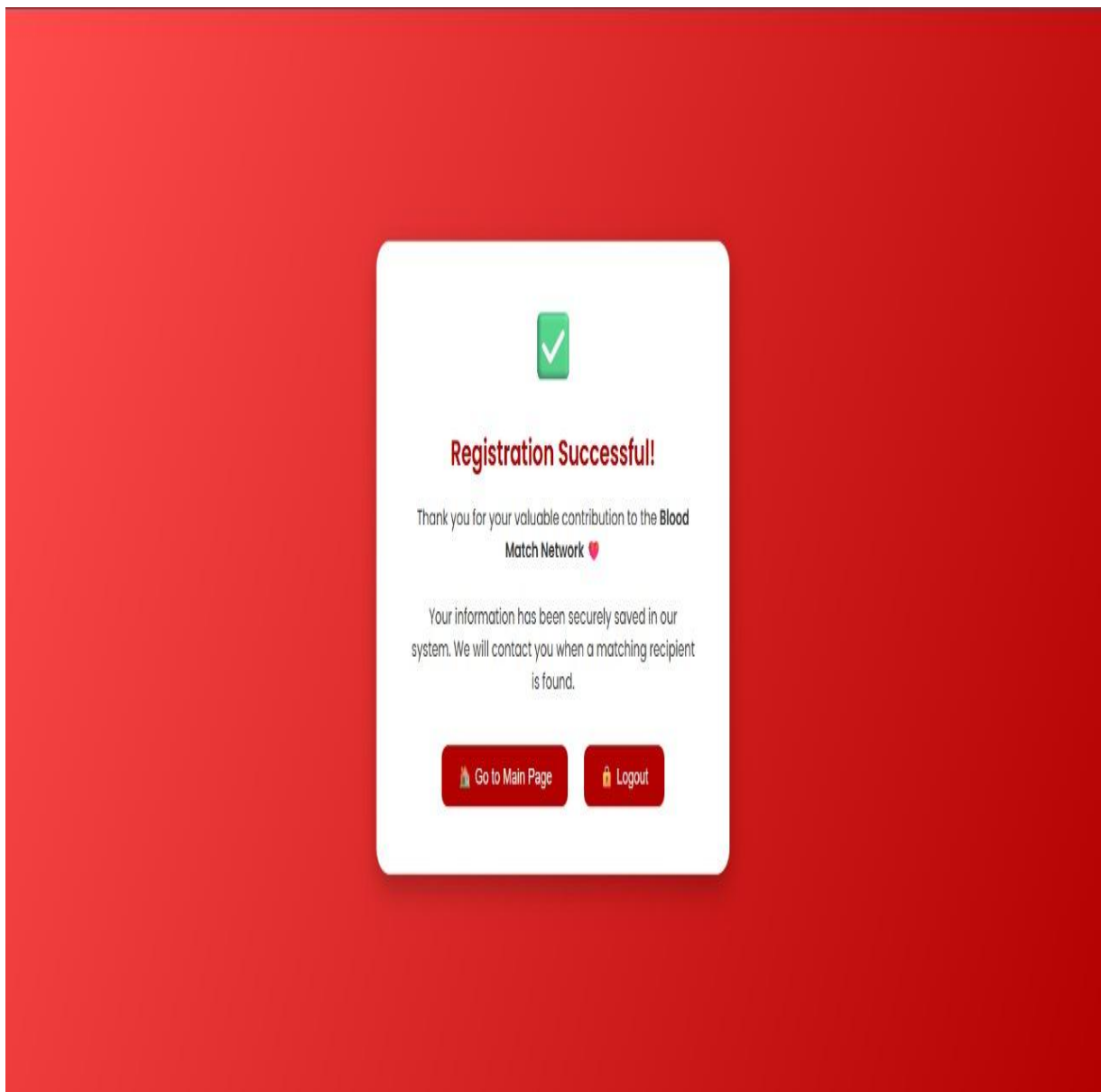


DONOR REGISTRATION FORM : This form is used to collect essential personal and medical details from a volunteer donor. It includes their demographics, contact info, blood group, and initial medical test results (like Platelet Count).

[illegible]

DONOR REGISTRATION SUCCESSFUL CONFIRMATION:

This is the final confirmation screen, assuring the donor that their contribution is valuable and their information is securely stored. The system promises to **contact them** when a recipient with a matching need is found.



MYSQL DATABENCH DONORS TABLE : This shows the back-end database structure where all the donor data is stored and managed. It proves that the front-end form data is being successfully captured into a structured table (DONORS) within the bloodmatch schema.

The screenshot displays the MySQL Workbench interface. On the left, the Navigator pane shows the 'bloodmatch' schema with a table named 'donors'. The central Query Editor shows the following SQL query:

```
1 • use bloodmatch;
2 • select * from donors;
```

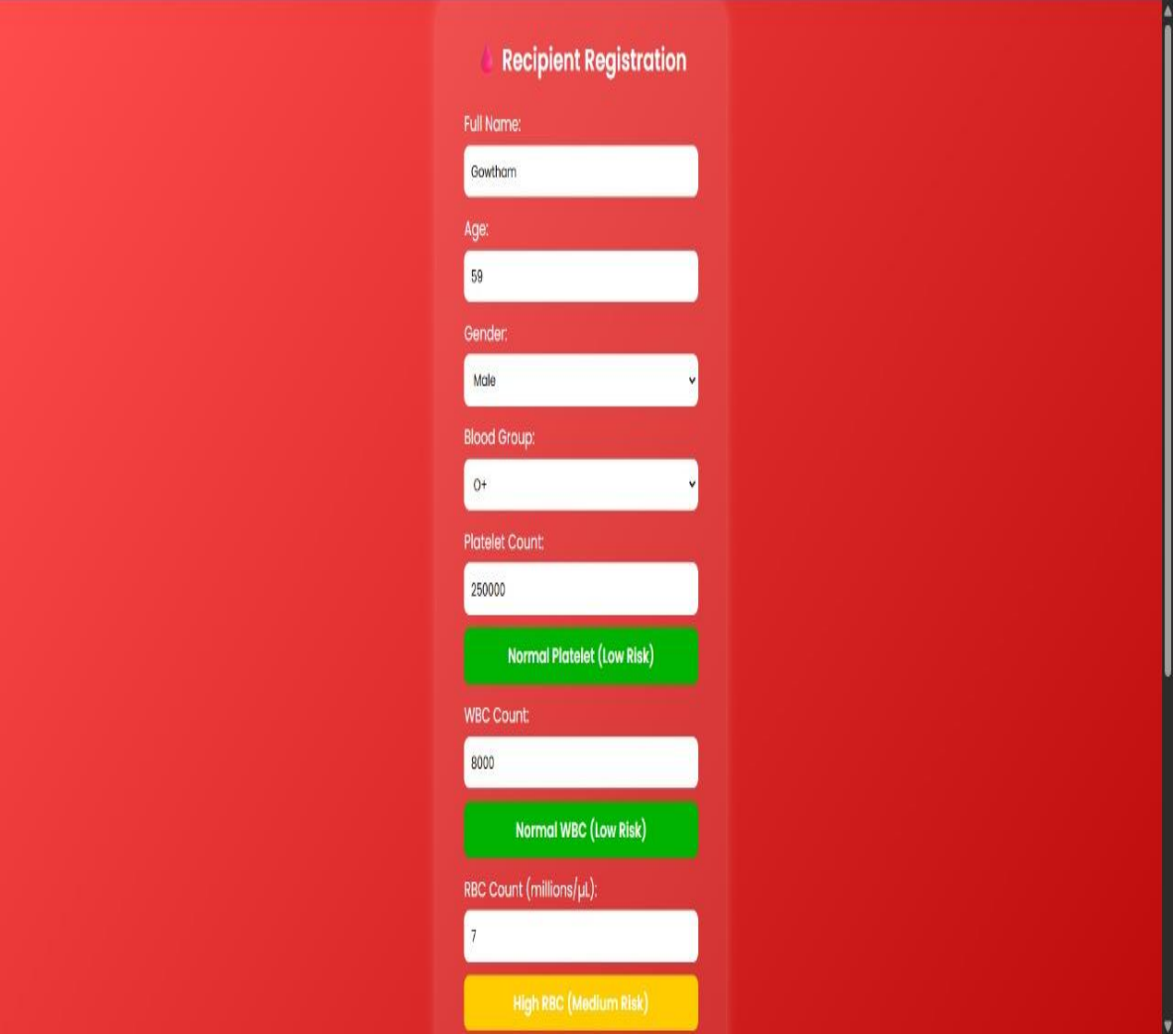
The Result Grid at the bottom displays the data for the 'donors' table. The table has 10 rows and 15 columns. The columns are: id, name, age, weight, bloodGroup, gender, contact, city, illness, illnessDetails, donatedRecently, plateletCount, rbc, vbc, hemoglobin, and riskFactor. The data is as follows:

id	name	age	weight	bloodGroup	gender	contact	city	illness	illnessDetails	donatedRecently	plateletCount	rbc	vbc	hemoglobin	riskFactor
1	Kathir	18	51	B+	Male	7200866675	chennai	Yes		Yes	250000	4.5	8000	12.3	High
3	Jai Suriya	18	80	O+	Male	9034234231	Coimbatore	No		No	250000	4.5	8000	12.3	High
4	Vinoth	19	55	B+	Male	1234567890	chennai	No		Yes	250000	4.5	8000	12.3	High
5	jana	19	52	O+	Female	7200866675	Coimbatore	No		No	250000	4.5	8000	12.3	High
6	Arun	20	55	B+	Male	7200866675	Coimbatore	No		No	250000	4.5	8000	12.3	High
7	venky	25	60	B+	Male	9453242234	mumbai	No		No	2452343	6	5000	12	Medium
8	thiru	23	70	AB+	Male	9453242234	Delhi	No		No	90000	4	7000	15	High
9	Harish	18	60	B+	Male	7200866675	chennai	No		No	260000	5	6000	15	Low
10	harshinee	20	55	B+	Female	7200381311	chennai	Yes		No	150000	4.5	8000	14.2	Low

The bottom pane shows the 'Output' tab with the following action log:

#	Time	Action	Message	Duration / Fetch
1	17:41:18	use bloodmatch	0 row(s) affected	0.000 sec
2	17:41:30	select * from donors LIMIT 0, 1000	9 row(s) returned	0.000 sec / 0.000 sec

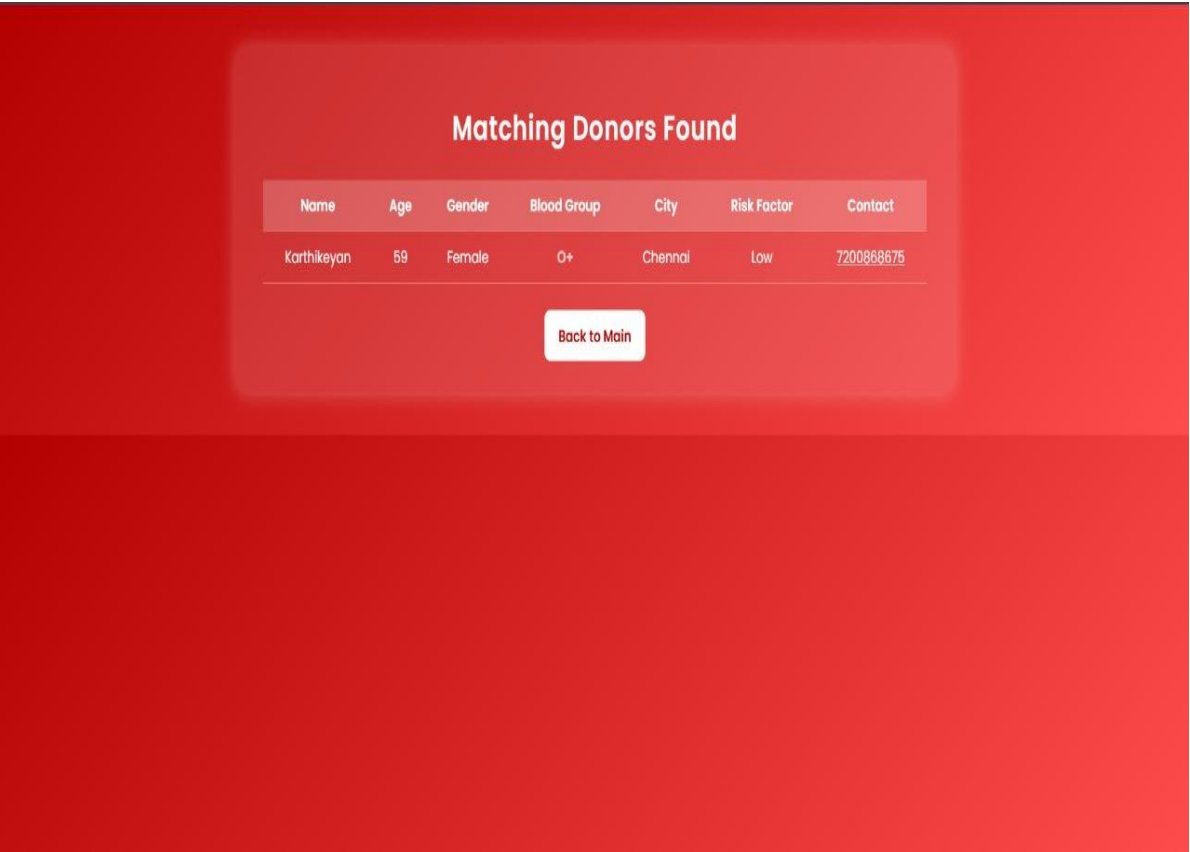
RECIPIENT REGISTRATION FORM : This part of the recipient form gathers detailed medical history, including Hemoglobin levels, major illnesses, and recent blood donation history. It helps the system accurately assess the recipient's risk and matching criteria



The image shows a web form titled "Recipient Registration" with a flame icon. The form is set against a red background. It contains several input fields for personal and medical information, each followed by a green or yellow button indicating a risk level.

Field	Value	Risk Level
Full Name:	Gowtham	
Age:	59	
Gender:	Male	
Blood Group:	O+	
Platelet Count:	250000	Normal Platelet (Low Risk)
WBC Count:	8000	Normal WBC (Low Risk)
RBC Count (millions/ μ l):	7	High RBC (Medium Risk)

DONOR RECIPIENT MATCHING : This is the results screen after a recipient's need is entered, showing potential matches (like 'Karthikeyan'). It provides the recipient with the donor's contact details, location, and risk assessment for follow-up.



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CONCLUSION & FUTURE ENHANCEMENT

In the future, the Blood Donation Management System can be enhanced by integrating a mobile application that allows donors and patients to easily register, request blood, and receive instant updates. Real-time blood stock monitoring across multiple hospitals can be added to improve availability and reduce delays during emergencies. Automated SMS and email alerts can notify donors about upcoming blood donation camps, urgent requirements, and their next eligible donation date. GPS-based donor location tracking can be introduced to quickly find nearby donors in critical situations. The system can also include online appointment booking to make the donation process faster and more organized. Additionally, AI can be used to match donors and recipients more accurately, while digital health records can help track donor eligibility and donation history. Finally, integrating block chain technology can make blood tracking more secure, transparent, and tamper-proof, improving trust and reliability in the overall system.

REFERENCES

Websites

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- <https://www.who.int> – World Health Organization guidelines on blood donation.
- <https://www.blooddonationcamp.com> – Awareness and database management insights.
- <https://www.geeksforgeeks.org> – Tutorials on Java, JDBC, and database connectivity.
- <https://www.javatpoint.com> – Java MySQL integration and DBMS tutorials.

Research Articles

- Sharma, P., & Gupta, R. (2020). *Blood Bank Management System Using Database and Web Technologies*. *International Journal of Computer Applications*, 176(36), 12–17.
- Kumar, S., & Raj, V. (2021). *Automation of Blood Donation Management System for Hospitals and Donor Centers*. *International Research Journal of Engineering and Technology (IRJET)*, 8(4), 1450–1454.

Software / Tools Used

- MySQL Workbench (Database Design & Query Execution)
- Eclipse (Java Programming & GUI)
- XAMPP (Local Server Environment)
- Draw.io / Lucidchart (ER Diagram Design)